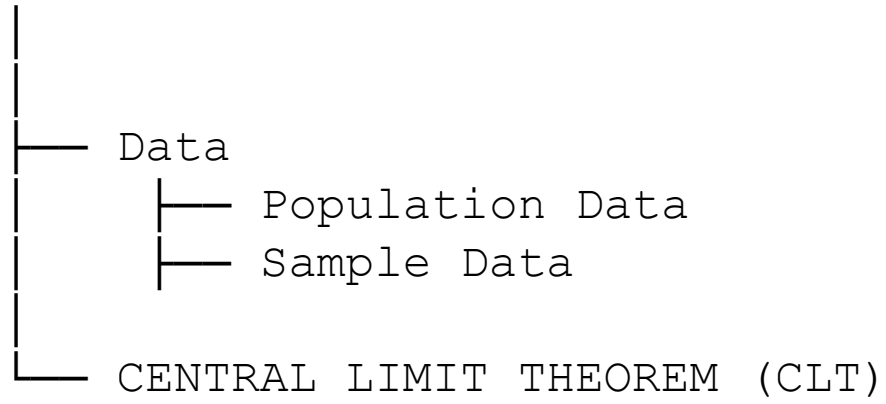


SAMPLING & SAMPLING DISTRIBUTIONS



Central Limit Theorem:

If you repeatedly take random samples of size n from any population and calculate the sample mean, the distribution of those sample means will approach a **normal distribution (bell curve)** as the sample size becomes large, regardless of the shape of the original population.

Number of Sample = m

Size of Sample = n

Mean of Each Sample

S_1 $\{X_1, X_2, X_3, X_4, \dots, X_n\}$ $\cdots \rightarrow \bar{X}_1$

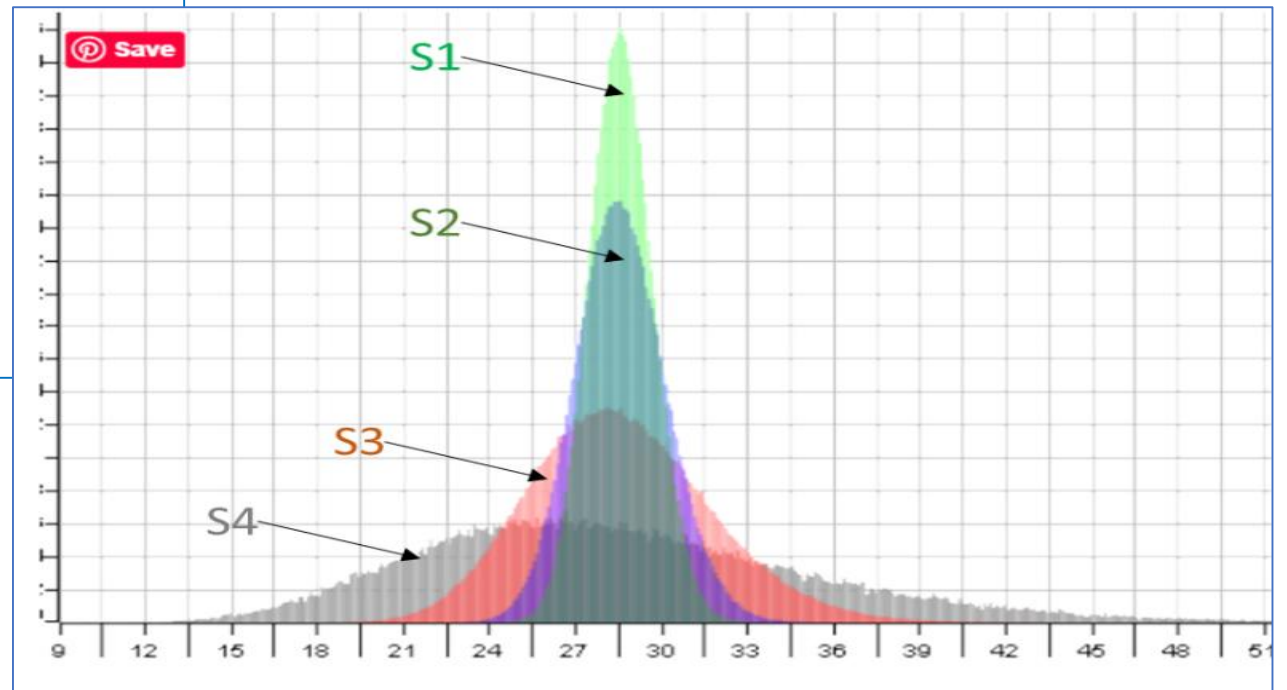
S_2 $\{X_1, X_2, X_3, X_4, \dots, X_n\}$ $\cdots \rightarrow \bar{X}_2$

S_3 $\{X_1, X_2, X_3, X_4, \dots, X_n\}$ $\cdots \rightarrow \bar{X}_3$

S_4 $\{X_1, X_2, X_3, X_4, \dots, X_n\}$ $\cdots \rightarrow \bar{X}_4$

$\vdots$

$\bar{X}_m$



1. Take many random samples of size 30.
2. Compute the mean of each sample.
3. Plot all those means.

The resulting distribution of means will look approximately like a bell curve (Normal Distribution).

What Happens to the Mean and Spread?

According to CLT:

Mean of sample means = population mean (μ)

Standard deviation of sample means =

This quantity is called the **Standard Error (SE)**.

As sample size increases:

Standard error decreases

Estimates become more precise

$$SE = \frac{\sigma}{\sqrt{n}}$$

← Standard deviation

← Number of samples